

1 Methodology

This study employs network science methods to analyze chess positions as directed graphs, computing six fundamental metrics—directed degree, closeness, and betweenness centralities; algebraic connectivity; size entropy; and directed modularity—across multiple temporal and structural scales. We develop a novel framework for component-level statistical pooling and node-level normalization to enable meaningful comparisons across heterogeneous network structures. The methodology extends established network science techniques ([2, 17]) to the domain of chess position analysis.

1.1 Graph Construction

We model each chess position as a directed graph $G = (V, E)$ following the theoretical framework established by Shannon ([21]) for game-tree complexity. The vertex set V comprises the 64 board squares, while the directed edge set E represents legal moves, where $(u \rightarrow v) \in E$ indicates a legal move from square u to square v . This representation extends graph-theoretic approaches to games ([18]) by focusing on positional structure rather than game-tree search.

For each position at ply t , we construct three directed graphs:

- **Combined influence graph** $G_C^{(t)}$: containing all legal moves by both players
- **White subgraph** $G_W^{(t)}$: containing only White’s legal moves
- **Black subgraph** $G_B^{(t)}$: containing only Black’s legal moves

Each vertex maintains attributes (occupancy status, piece type, color) following standard chess notation. Edge construction employs the `python-chess` library for legal move generation and NetworkX ([11]) for graph representation. Isolated vertices (nodes with zero in-degree and out-degree) are removed to focus analysis on the active network structure.

1.2 Network Metrics

1.2.1 Directed Degree Centrality

Following Freeman’s ([10]) foundational framework for centrality measures in directed networks, we compute in-degree and out-degree centralities for each node v within its weakly connected component of size n_k :

$$C_{\text{in}}(v) = \frac{\deg^-(v)}{n_k - 1}, \quad C_{\text{out}}(v) = \frac{\deg^+(v)}{n_k - 1} \quad (1)$$

where $\deg^-(v)$ and $\deg^+(v)$ denote the in-degree and out-degree of node v , respectively. This normalization by $(n_k - 1)$ represents the maximum possible edges within the component ([24, Ch. 5]).

1.2.2 Directed Closeness Centrality

We implement directed closeness centrality following Sabidussi ([19]) and Freeman ([10]), distinguishing between out-closeness (influence spreading) and in-closeness (reachability):

$$C_{\text{out}}(v) = \frac{n_k - 1}{\sum_{u \neq v} d(v, u)}, \quad C_{\text{in}}(v) = \frac{n_k - 1}{\sum_{u \neq v} d(u, v)} \quad (2)$$

where $d(x, y)$ denotes the directed shortest path length from x to y . For disconnected components, we adopt the harmonic mean formulation ([15]) to handle infinite distances.

1.2.3 Directed Betweenness Centrality

Directed betweenness centrality quantifies the extent to which node v lies on shortest paths between other node pairs ([9]):

$$g(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}} \quad (3)$$

where σ_{st} is the number of shortest paths from s to t , and $\sigma_{st}(v)$ is the number of those paths passing through v . We normalize by $(n_k - 1)(n_k - 2)$ for directed graphs within components of size n_k ([5]). Implementation uses Brandes’ algorithm with $O(nm)$ complexity for unweighted graphs.

1.2.4 Algebraic Connectivity (Fiedler Value)

For each weakly connected component, we compute the algebraic connectivity as the second-smallest eigenvalue λ_2 of the undirected Laplacian matrix $L = D - A$, where D is the degree matrix and A is the adjacency matrix ([8]). This metric quantifies the component's robustness to disconnection and relates to graph expansion properties ([6]).

1.2.5 Size Entropy

We introduce size entropy to characterize the distribution of weakly connected component sizes. Let $\{n_k\}$ denote component sizes with probabilities $p_k = n_k / \sum_j n_j$. The size entropy:

$$H = - \sum_k p_k \ln p_k \quad (4)$$

ranges from near 0 (single dominant component) to $\ln K$ (uniform distribution across K components). This measure extends Shannon's ([20]) information entropy to network component analysis, analogous to applications in network ensemble theory ([1]).

1.2.6 Directed Modularity

We employ directed modularity ([14]) to detect community structure:

$$Q = \frac{1}{m} \sum_{ij} \left[A_{ij} - \frac{k_i^{\text{out}} k_j^{\text{in}}}{m} \right] \delta(c_i, c_j) \quad (5)$$

where A_{ij} is the adjacency matrix, k_i^{out} and k_j^{in} are out- and in-degrees, m is the total number of edges, and $\delta(c_i, c_j) = 1$ if nodes i and j belong to the same community. We optimize using the Leiden algorithm ([23]), which provides mathematical guarantees for well-connected communities.

1.3 Component Decomposition and Statistical Pooling

Following established methods for multi-component networks ([13]), we decompose each graph into weakly connected components using depth-first search ([22]). For component k with size n_k , mean μ_k , and variance σ_k^2 of a given metric, we compute global statistics using size-squared weighting:

$$\mu_{\text{global}} = \frac{\sum_k n_k^2 \mu_k}{\sum_k n_k^2} \quad (6)$$

$$\sigma_{\text{global}}^2 = \frac{\sum_k n_k^2 \sigma_k^2}{\sum_k n_k^2} + \frac{\sum_{i < j} n_i^2 n_j^2 (\mu_i - \mu_j)^2}{(\sum_k n_k^2)^2} \quad (7)$$

This weighting scheme reflects the $O(n_k^2)$ scaling of path-based metrics and provides unbiased estimates analogous to weighted variance in meta-analysis ([7]).

1.4 Normalization Invariants

Two key invariants validate our computations:

1. **Degree handshake theorem:** For directed graphs, $\sum_v \deg^-(v) = \sum_v \deg^+(v) = |E|$ ([25]).
2. **Closeness variance symmetry:** The variances of in-closeness and out-closeness are equal due to the mathematical property that computing in-closeness on graph G is equivalent to computing out-closeness on the transpose graph G^T (where all edge directions are reversed). This follows directly from the definition of directed closeness centrality ([10]).

1.5 Node-Level Centrality Scaling

To enable comparison across components of different sizes, we rescale centrality measures to a common $[0, 1]$ range while preserving within-component ordering ([4]):

For degree and closeness centralities:

$$\tilde{C}(v) = C_k(v) \times \frac{n_k - 1}{N - 1} \quad (8)$$

For betweenness centrality:

$$\tilde{g}(v) = g_k(v) \times \frac{(n_k - 1)(n_k - 2)}{(N - 1)(N - 2)} \quad (9)$$

where N is the total number of nodes in the full graph and $C_k(v)$ denotes the component-normalized centrality.

1.6 Dynamic Time-Series Analysis

We perform longitudinal analysis following established methods for temporal networks ([12]). For each game represented in Portable Game Notation (PGN), we iterate through plies $t = 0, 1, 2, \dots$ and execute:

1. Construct directed graphs $G_C^{(t)}$, $G_W^{(t)}$, and $G_B^{(t)}$
2. Compute component-level metrics for each weakly connected component
3. Aggregate to global metrics using size-squared pooling (Section 3.3)
4. Scale node-level centralities to global normalization (Section 3.5)
5. Record time series of all metrics

This approach extends static network analysis to capture dynamic evolution of positional structure ([3]).

1.7 Interpretation Framework Based on Size Entropy

We employ size entropy as a structural fingerprint to guide comparative analysis:

- **Low entropy** ($H < 0.3$): Single dominant component; aggregate metrics are representative
- **High entropy** ($H > 0.8$): Multiple similarly-sized components; requires component-level analysis
- **Entropy difference** ($\Delta H > 0.2$): Structurally different graphs; use percentile-based ranking or union graph analysis

This framework prevents misinterpretation of aggregate statistics when underlying structures differ fundamentally, addressing known issues in multi-component network comparison ([16]).

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